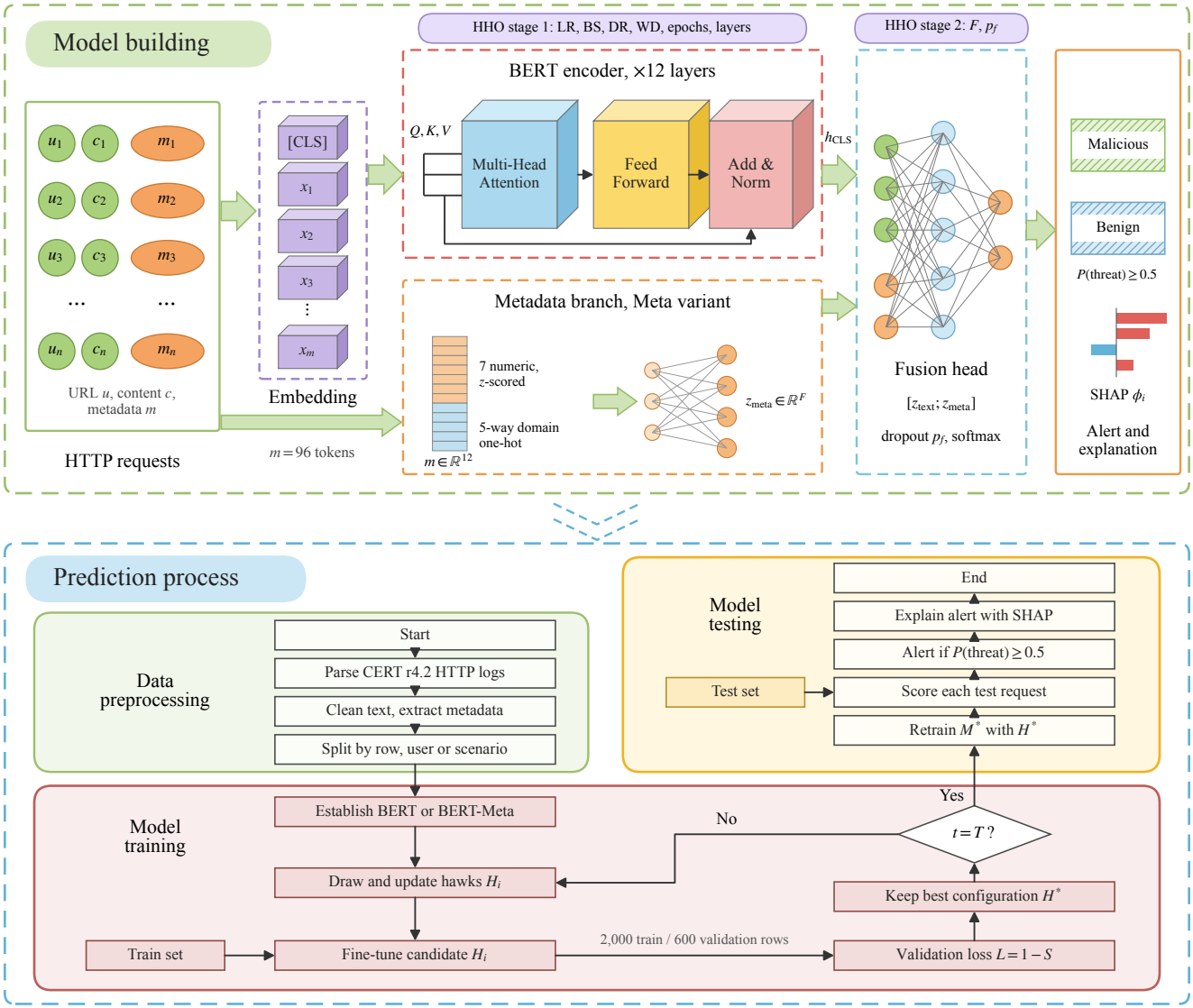

Model building
HHO stage 1: LR, BS, DR, WD, epochs, layers
HHO stage 2: F, p_f
BERT encoder, ×12 layers
Q, K, V
Multi-Head Attention
Feed Forward
Add & Norm
h_{CLS}
u_1 c_1 m_1
u_2 c_2 m_2
u_3 c_3 m_3
u_n c_n m_n
URL u, content c, metadata m
HTTP requests
[CLS]
x_1
x_2
x_3
x_m
Embedding
$m = 96$ tokens
Metadata branch, Meta variant
7 numeric, z-scored
5-way domain one-hot
$m \in \mathbb{R}^{12}$
$z_{meta} \in \mathbb{R}^F$
Fusion head
$[z_{text}; z_{meta}]$
dropout p_f, softmax
Malicious
Benign
$P(\text{threat}) \geq 0.5$
SHAP ϕ_i
Alert and explanation
Prediction process
Data preprocessing
Start
Parse CERT r4.2 HTTP logs
Clean text, extract metadata
Split by row, user or scenario
Model training
Establish BERT or BERT-Meta
Draw and update hawks H_i
Train set
Fine-tune candidate H_i
2,000 train / 600 validation rows
Validation loss $L = 1 - S$
Keep best configuration H^*
$t = T$?
No
Yes
Model testing
Test set
Score each test request
Alert if $P(\text{threat}) \geq 0.5$
Explain alert with SHAP
End
Retrain M^* with H^*